

Digital Clock: Progress Log

B.Tarun Tej & B.Rajaswanik

August 28, 2026

1 Contents

1. Introduction
2. Materials and their principle
3. Building and Working
4. code and its Working
5. Problems Faced and their solutions
- 6.

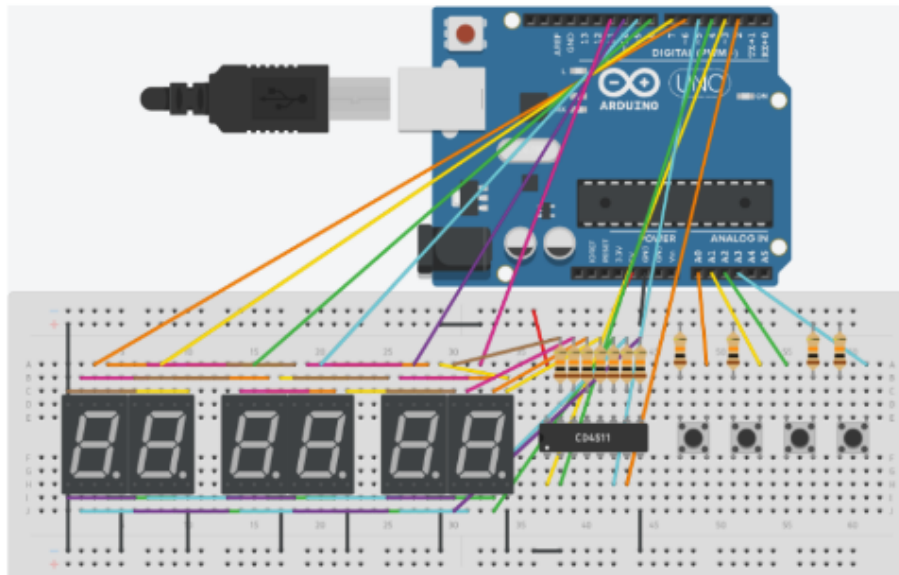


Figure 1: Tinker-Card simulation of the project(taken from sirs pdf)

2 Introduction

The digital clock is based on basic principles of digital electronics and arduino programming, in this project we use truth table and use kmaps for its implementation in order to use only one ic 7447 we use multiplexing method we also do cpp programing in this project

3 Materials and their principle

1.Arduino uno The Arduino Uno is an open-source microcontroller board widely used for learning electronics, rapid prototyping, and embedded systems design. There are different types of pins in the arduino uno like power pins,digital pins,analog pins and some digital pins can control power



Figure 2: Arduino Uno

2. **7 segment anode display** A seven-segment display (SSD) is an electronic device used to present numerical digits (0–9) and a limited set of alphanumeric characters. It consists of seven individual illuminating bar segments arranged in a figure-8 layout, plus an optional eighth segment for a decimal point (DP).

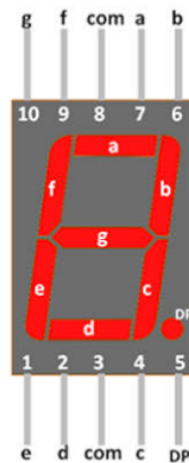


Figure 3: Seven Segment Display(anode)

3.Bread Board: A breadboard (or solderless breadboard) is a construction base used for prototyping temporary electronic circuits without soldering. Inside the plastic housing are spring-metal clips that hold component leads and wires in place while keeping them electrically connected.

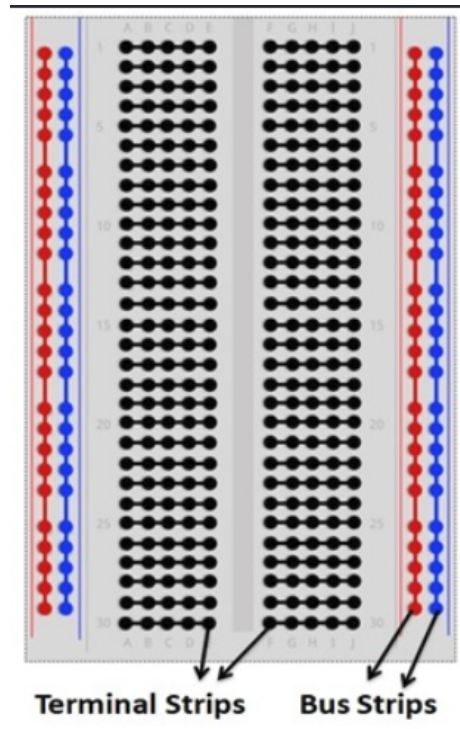


Figure 4: Bread Board.

4.miscellaneous:

1. jumper wires male to male
2. solid core wires
3. resistors
4. push buttons



Figure 5: jumper wires.

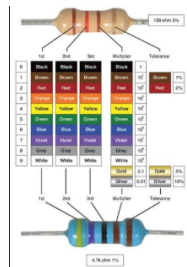


Figure 6: Resistor.



Figure 7: Push Buttons

4 Building and Working

Building the Clock

Component List

Microcontroller: 1 Arduino Uno with Type B USB Cable

Display: 6 Common Cathode Seven-Segment

Decoder: 1 IC 7447 BCD-to-7-Segment

Inputs Passive Components: 4 Push Buttons, 16 Male-to-Male Jumper Wires, 1 Breadboard, 7 220 Ω resistors, and 4 10 k Ω pull-down resistors

Wiring

IC 7447 BCD-to-7Segment Inputs: Connect the BCD input Pins 7, 1, 2, and 6 of the 7447 IC to the Arduino digital pins D0, D1, D2, and D3.

Multiplexing Display Pins: Connect the Arduino pins D4 – D9 to the Enable pins EN[0..5] of the display driver (EN[0..5] are for the 6 displays: Hours Tens/Units, Minutes Tens/Units, Seconds Tens/Units).

Segments Display Pins: The common cathode pins a – g of the display should be parallel-connected from the 7447 IC with resistor voltage limiting to the 6 displays (only 7 segment output wires in total), with the 7447 Pin 8 to GND and Pin 16 to 5V.

Push Buttons Inputs: Connect the four Push Buttons (with 10 k Ω pull-down resistors) to the Arduino pins D10(Edit/Pause Toggle), D11(Next Digit), D12 (Increment), and D13 (Decrement).

Working Principle

Display Multiplexing Working Principle: All 6 displays share the same BCD bus. By rapidly switching on/off (1 ms) all displays with different Enable pins connected to Arduino pins D4-D9, the human eye perceives a continuous image due to persistence of vision.

Clock Ticking Working Principle: Time-keeping is managed by a 10 Hz timer interrupt. The time variables are properly updated, considering only valid intervals for seconds, minutes (00 – 59), and hours (00 – 23) in a 24-hour format.

Editing Working Principle: By pressing the first button, the user can switch between run/edit mode, and pause time-tracking in the edit mode.

Digit Choose Working Principle: The second button cycles through the editable digits in the following order: sec ones $\rightarrow$ sec tens $\rightarrow$ min ones $\rightarrow$ min tens $\rightarrow$ hr ones $\rightarrow$ hr tens). The selected digit blinks every 200 ms to indicate it is editable.

Value Change Working Principle: The third and fourth buttons change the value of the currently selected digit, following the Boolean logic obtained from Karnaugh maps to increment/decrement the binary value and manage carry-over operations (rollover) and decrement (underflow) transitions while maintaining valid time intervals.

5 code and its working

The code is in `digital_clock_code.ino` This is the code for 24 hour clock

5.1 Hardware setup and Multiplexing

Data Output $-i$ gives binary bits (A,B,C,D) to IC 7447 Enable Control $-i$ controls multiplexing enabling it and making so that which display is turned on for the particular instant Control buttons $-i$ these are for setting time void setup works one time and setup the pins the pin mode function sets up the pins of the arduino void loop function works repeatedly until the arduino stops

5.2 K-maps Logic tables

Z is taken as 2^3 part in the binary in the input part

Y is taken as 2^2 part in the binary in the input part

X is taken as 2^1 part in the binary in the input part

W is taken as 2^0 part in the binary in the input part

for seconds ones digit logic subscript 1 is used for seconds tens digit logic subscript 2 is used and so on similarly

D is taken as 2^3 part in the binary in the output part

C is taken as 2^2 part in the binary in the output part

B is taken as 2^1 part in the binary in the output part

A is taken as 2^0 part in the binary in the output part

for seconds one and minutes one (0-9) is boundary conditions for seconds ones and minutes ones (0-5) is boundary condition

millis() function is used to track time elapsed it acts as base 1 - second interval trigger

The buttons do what they are named after exactly millis() function is paused while setting the time

6 Problems Faced and thier solutions

6.1 Displays mixup

some displays given to us were cathode and some were anode when we connected them to the bread board the displays were some displays were not working and some of them were only working partially

solution found out the difference between anode and cathode displays verified them each sepratley and connected only anode displays to the bread board

6.2 Displays partially not working

even after connecting the displays and multiplexing and connecting all of the some of the segments of seven segment display were not lighting up properly

solution changed the bread board doubting its condition due to visible damages on it luckily it worked and the displays started working

6.3 usage of jumper wires for multiplexing

we found using jumper wires for multi plexing looked and felt messy so we learned that we can use solid core wires instead of them for mutiplexing now it kind of looks organized and definetly looks way better

6.4 The solid core wire broke

one of the solid core wire broke making display segments not working properly again had to do all of the multiplexing again to find where the problem occured

7 Digital Clock

The picture down here is our final build of the digital clock showing the time in 24 hour format

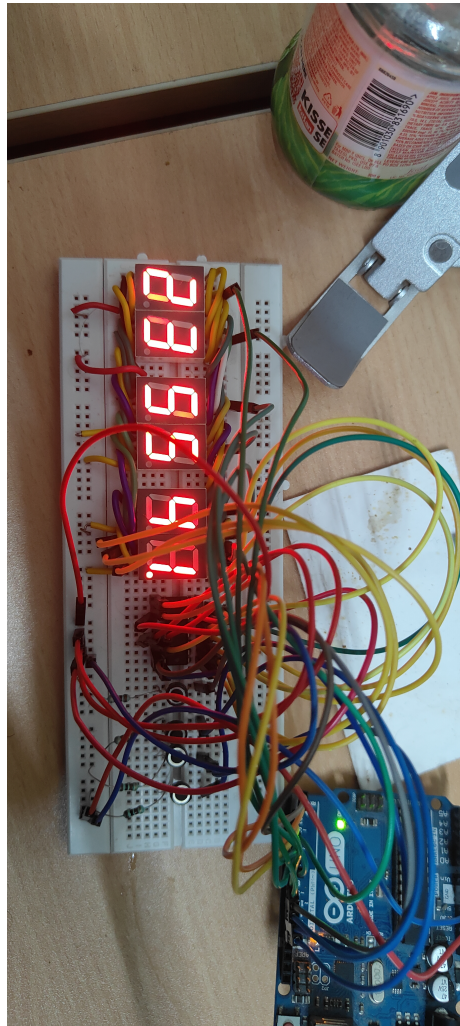


Figure 8: A sample chart illustrating our findings.